

chapter-3 Review of Z-transform

The z-transform is the discrete time analysis of the Laplace transform. The z-transform of a discrete-time signal $x(n)$ is defined as the power series,

$$X(z) = Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \quad \text{--- (1)}$$

where z is complex variable.

Eqⁿ (1) is sometimes called the direct z-transform because it transforms $x(n)$ into complex-plane representation $X(z)$.

The inverse procedure [i.e. obtaining $x(n)$ from $X(z)$] is called the inverse z-transform i.e.

$$x(n) = z^{-1}[X(z)]$$

The relationship between $x(n)$ & $X(z)$ is indicated by

$$x(n) \xleftrightarrow{Z} X(z)$$

Since, the z-transform is an infinite power series, it exists only for those values of z for which the series converges. The region of convergence (ROC) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.

Thus, any time we cite a z-transform, we should also indicate its ROC.

Advantages of z-transform:

1. Discrete time signals and LTI systems can be completely characterized by z-transform.
2. The stability of LTI system can be determined using z-transform.
3. By calculating z-transform of given signal, DFT and FT can be determined.
4. The solution of differential equations can be simplified using z-transform.
5. Mathematical calculations are reduced using z-transform. e.g. Convolution operation is transformed into simple multiplication operation.
6. Entire family of digital filters can be obtained from one proto-type design using z-transform.

one sided z-transform:

A single side z-transform of discrete time signal $x(n)$ may be defined as

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$\text{OR} \quad X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

which limits the summation from 0 to ∞ . while expanding we put only positive values of n (i.e. $n=0$ to $n=\infty$). Due to this, the one-sided z-transform has the following characteristics.

- i) It does not contain ^{any} information about the signal $x(n)$ for negative values of time (i.e. for $n < 0$)

ii) It is unique only for causal signals, because only these signals are zero for $n < 0$.

iii) The one-sided z-transform $X^+(z)$ of $x(n)$ is identical to two-sided z-transform of the signal $x(n)u(n)$, the ROC of $X^+(z)$ is always the exterior of a circle.

b) Two-sided z-transform:

A double sided z-transform of discrete-time signal $x(n)$ may be defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

While expanding the summation, we shall put both positive & negative values of 'n'.

Region of Convergence (ROC)

The set of values of z in the z -plane for which the magnitude of $X(z)$ is finite is called the region of convergence (ROC).

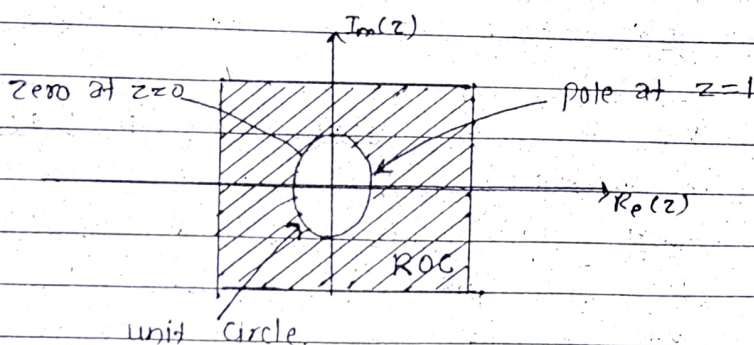


Fig: pole-zero plot and ROC of unit step discrete time signal $u(n)$

Properties of ROC:

1. The ROC of $X(z)$ consists of a ring circle in the z -plane centered about the origin.

2. The ROC does not contain any pole.

3. The z -transform $X(z)$ converges uniformly if and only if the ROC of $X(z)$ includes the unit circle.
i.e. ROC of $X(z)$ has value of z for which

$$\sum_{n=-\infty}^{\infty} |x(n) r^n| < \infty$$

4. If $x(n)$ is a finite duration, then the ROC will be the entire z -plane except $z=0$ and $z=\infty$.

5. If $x(n)$ is Right Sided sequence, then ROC will not include infinity.

6. If $x(n)$ is left-sided sequence, then ROC will not include $z=0$. But in a case if $x(n)=0$ for all $n>0$, then the ROC will include $z=0$.

7. If $x(n)$ is a two sided sequence, then ROC will ~~not~~ include consists of a ring in z -plane that includes the circle $|z|=r_0$ i.e. ROC will include the intersection of ROC's components.

8. If $X(z)$ is rational, then ROC will extend to infinity.

9. If $x(n)$ is causal, then ROC will include $z=\infty$.

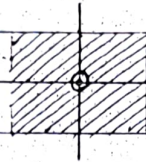
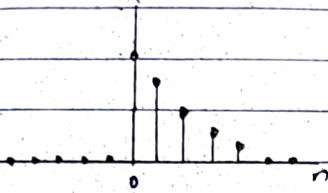
10. If $x(n)$ is anti-causal, then ROC will include $z=0$.

Signal

ROC

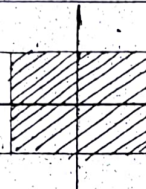
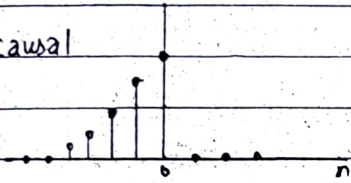
Finite - Duration Signals

Causal



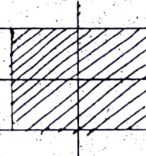
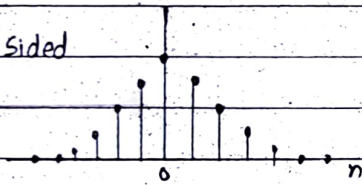
Entire z-plane
except $z=0$

Anticausal



Entire z-plane
except $z=\infty$
and

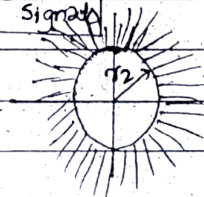
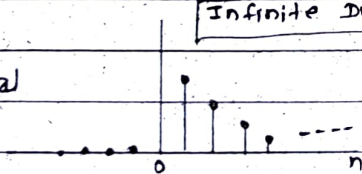
Two-Sided



Entire z-plane
except $z=0$ and
 $z=\infty$

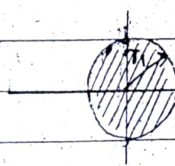
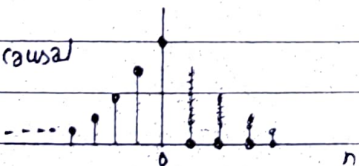
Infinite Duration Signals

Causal



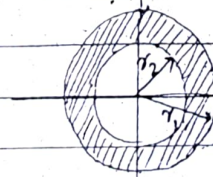
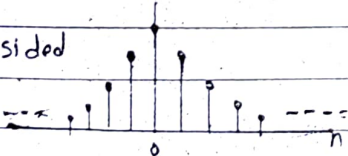
$|z| > r_2$

Anti-causal



$|z| < r_1$

Two-sided



$r_2 \leq |z| < r_1$

properties of z-transform:

1. Linearity:

$$\text{If } x_1(n) \xrightarrow{z} X_1(z)$$

$$\text{and } x_2(n) \xrightarrow{z} X_2(z) \text{ then}$$

$$x(n) = a_1 x_1(n) + a_2 x_2(n) \xrightarrow{z} X(z) = a_1 X_1(z) + a_2 X_2(z)$$

where

a_1 & a_2 are any constant.

2. Time Shifting:

$$\text{If } x(n) \xrightarrow{z} X(z) \text{ then}$$

$$x(n-k) \xrightarrow{z} z^{-k} X(z)$$

Roc of $z^{-k} X(z)$ is the same as that of $X(z)$ except for $z=0$ if $k > 0$ and $z=\infty$ if $k < 0$.

3. Scaling in z-domain:

$$\text{If } x(n) \xrightarrow{z} X(z) \quad \text{Roc: } r_1 < |z| < r_2$$

then,

$$a^n x(n) \xrightarrow{z} X(a^{-1}z) \quad \text{Roc: } |a| r_1 < |z| < |a| r_2$$

where, a is any constant may be real or complex

4. Time reversal: (affect Roc)

$$\text{If } x(n) \xrightarrow{z} X(z) \quad \text{Roc: } r_1 < |z| < r_2$$

then

$$x(-n) \xrightarrow{z} X(z^{-1}) \quad \text{Roc: } \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

5. Differentiation in z-domain:

If $x(n) \xleftrightarrow{Z} X(z)$
 then, } Same ROC

$$n x(n) \xleftrightarrow{Z} -z \frac{dX(z)}{dz}$$

$$\text{or } n x(n) \xleftrightarrow{Z} -z^{-1} \frac{dX(z)}{dz^{-1}}$$

6. Convolution of two sequences:

If $x_1(n) \xleftrightarrow{Z} X_1(z)$

and

$x_2(n) \xleftrightarrow{Z} X_2(z)$ then,

$$x(n) = x_1(n) * x_2(n) \xleftrightarrow{Z} X(z) = X_1(z) X_2(z)$$

7. Multiplication of two sequences:

If $x_1(n) \xleftrightarrow{Z} X_1(z)$

and

$x_2(n) \xleftrightarrow{Z} X_2(z)$ then

$$x(n) = x_1(n) x_2(n) \xleftrightarrow{Z} X(z) = \frac{1}{2\pi j} \oint_C X_1(v) X_2\left(\frac{z}{v}\right) v^{-1} dv$$

8. Correlation of two sequences:

If $x_1(n) \xleftrightarrow{Z} X_1(z)$

and

$x_2(n) \xleftrightarrow{Z} X_2(z)$ then,

$$r_{x_1, x_2}(l) = \sum_{n=-\infty}^{\infty} x_1(n) x_2(n-l) \xleftrightarrow{Z} R_{x_1, x_2}(z) = X_1(z) X_2(z^{-1})$$

9. Initial value Theorem:

If $x(n)$ is causal [i.e. $x(n)=0$ for $n < 0$] then initial value of $x(n)$ is

$$x(0) = \lim_{n \rightarrow 0} x(n) = \lim_{|z| \rightarrow \infty} X(z)$$

10. Final value Theorem

For a discrete time signal $x(n)$, if $X(z)$ and poles of $X(z)$ are inside the unit circle, then the final value of $x(n)$ is

$$x(\infty) = \lim_{n \rightarrow \infty} x(n) = \lim_{|z| \rightarrow 1} [(1-z^{-1}) X(z)]$$

11. Parseval's Theorem:

If $x_1(n)$ and $x_2(n)$ are complex-valued sequences, then

$$\sum_{n=-\infty}^{\infty} x_1(n) x_2^*(n) = \frac{1}{2\pi j} \oint_C X_1(v) X_2^* \left(\frac{1}{v^*} \right) v^{-1} dv$$

provided that $\gamma_{1L} \gamma_{2L} < 1 < \gamma_{1U} \gamma_{2U}$, where, $\gamma_{1L} < |z| < \gamma_{1U}$ and $\gamma_{2L} < |z| < \gamma_{2U}$ are ROC of $X_1(z)$ and $X_2(z)$.

Inverse z-transform:

The procedure for transforming from z-domain to the time domain is called the inverse z-transform. Mathematically,

$$x(n) = z^{-1} [X(z)]$$

To perform inverse z-transform, any one of these methods are used.

- i) Long division method / power series expansion
- ii) Partial fraction expansion
- iii) Residue method.

i) Long division Method:

The z-transform of $x(n)$ is

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

which may be expressed as the ratio of two polynomials in z ,

$$X(z) = \frac{N(z)}{D(z)}$$

The above ratio of polynomials for z-transform may be divided out to produce a power series in the form of an equation with the coefficients representing the sequence values in the time domain as

$$X(z) = \frac{N(z)}{D(z)} = \sum_{n=0}^{\infty} a_n z^{-n} = a_0 z^0 + a_1 z^{-1} + a_2 z^{-2} + \dots$$

where coeff. a_n are the values of $x(n)$

2. Partial fraction method:

With the help of partial fraction method, we convert system transfer function into a sum of standard functions.

To do this, firstly denominator of the transfer function $H(z)$ is factorized into prime factors and then simple poles of $H(z)$ is obtained.

$H(z)$ is given by

$$H(z) = A_0 + \frac{A_1}{z-p_1} + \frac{A_2}{z-p_2} + \dots + \frac{A_N}{(z-p_N)}$$

3. Residue method:

In this method, we obtain inverse z-transform $x(n)$, by summing residues of $[X(z) z^{n-1}]$ at all poles, mathematically

$$x(n) = \sum_{\text{all poles of } X(z)} \text{residues of } [X(z) z^{n-1}]$$

The residue for any pole of order m at $z = \beta$ is

$$\text{Residue} = \frac{1}{(m-1)!} \lim_{z \rightarrow \beta} \left\{ \frac{d^{m-1}}{dz^{m-1}} [(z-\beta)^m X(z) \cdot z^{n-1}] \right\}$$

Signal $x(n)$	z-transform $X(z)$	Roc
1. $\delta(n)$	1	Entire z-plane
2. $\delta(n-k)$	z^{-k}	Entire z-plane except $z=0$.
3. $\delta(n+k)$	z^k	Entire z-plane except $z=\infty$
4. $u(n)$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$ z > 1$
5. $u(-n)$	$\frac{z^{-1}}{z^{-1}-1}$	$ z > 1$
6. $n u(n)$	$\frac{z}{(z-1)^2}$	$ z > 1$
7. $a^n u(n)$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$ z > a $

Signal	z-transform	Roc.
8. $n a^n u(n)$	$\frac{a z^{-1}}{(1 - a z^{-1})^2}$	$ z > a $
9. $-a^n u(-n-1)$	$\frac{1}{1 - a z^{-1}}$	$ z < a $
10. $-n a^n u(-n-1)$	$\frac{a z^{-1}}{(1 - a z^{-1})^2}$	$ z < a $
11. $(\cos \omega_0 n) u(n)$	$\frac{1 - z^{-1} \cos \omega_0}{1 - 2 z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
12. $(\sin \omega_0 n) u(n)$	$\frac{z^{-1} \sin \omega_0}{1 - 2 z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
13. $(a^n \cos \omega_0 n) u(n)$	$\frac{1 - a z^{-1} \cos \omega_0}{1 - 2 a z^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
14. $(a^n \sin \omega_0 n) u(n)$	$\frac{a z^{-1} \sin \omega_0}{1 - 2 a z^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

Additional property of ^{one side} z-transform: (shifting)

1. Time delay

If $x(n) \xrightarrow{z^+} X^+(z)$ then

$$x(n-k) \xrightarrow{z^+} z^{-k} \left[X^+(z) + \sum_{n=1}^k x(-n) z^n \right], \quad k > 0,$$

In case, $x(n)$ is causal, then

$$x(n-k) \xrightarrow{z^+} z^{-k} X^+(z)$$

proof:

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$z^+ \{x(n-k)\} = z^{-k} \left[\sum_{l=-k}^{-1} x(l) z^{-l} + \sum_{l=0}^{\infty} x(l) z^{-l} \right]$$

$$= z^{-k} \left[\sum_{l=-1}^{-k} x(l) z^{-l} + X^+(z) \right]$$

2. Time advance

If $x(n) \xrightarrow{z^+} X^+(z)$ then

$$x(n+k) \xrightarrow{z^+} z^k \left[X^+(z) - \sum_{n=0}^{k-1} x(n) z^{-n} \right], \quad k > 0 \quad \text{--- (2)}$$

proof:

$$z^+ \{x(n+k)\} = \sum_{n=0}^{\infty} x(n+k) z^{-n} = z^k \cdot \sum_{l=k}^{\infty} x(l) z^{-l}$$

where, we have changed the index of summation from n to $l = n+k$. then

$$X^+(z) = \sum_{l=0}^{\infty} x(l) z^{-l} = \sum_{l=0}^{k-1} x(l) z^{-l} + \sum_{l=k}^{\infty} x(l) z^{-l}$$

By combining we get the desired result eqn (2).

find the inverse

$x(n) = \{1, 2, 3\}$ & $y(n) = \{3, 4, 8, 8, 3\}$, find $h(n) = ?$

$$X(z) = 1 + 2z^{-1} + 3z^{-2} \quad \& \quad Y(z) = 3 + 4z^{-1} + 8z^{-2} + 8z^{-3} + 3z^{-4}$$

$$H(z) = \frac{Y(z)}{X(z)}$$

$$\begin{array}{r} 3 - 2z^{-1} + 3z^{-2} \\ 1 + 2z^{-1} + 3z^{-2} \overline{) 3 + 4z^{-1} + 8z^{-2} + 8z^{-3} + 3z^{-4}} \\ \underline{3 + 6z^{-1} + 9z^{-2}} \phantom{+ 8z^{-3} + 3z^{-4}} \\ -2z^{-1} - z^{-2} + 8z^{-3} \phantom{+ 3z^{-4}} \\ \underline{-2z^{-1} - 4z^{-2} - 6z^{-3}} \phantom{+ 3z^{-4}} \\ + \phantom{-2z^{-1} -} + \phantom{-2z^{-1} -} + \phantom{-2z^{-1} -} \phantom{+ 3z^{-4}} \\ 3z^{-2} + 14z^{-3} + 3z^{-4} \\ \underline{3z^{-2} + 6z^{-3}} \phantom{+ 3z^{-4}} \\ \phantom{3z^{-2} +} 8z^{-3} + 3z^{-4} \end{array}$$

2014-spring

2.b) Soln

Determine the causal signal $x(n]$ having $X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})^2}$

$$\text{Here, } X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})^2} = \frac{1}{(1-2z^{-1})} \cdot \frac{1}{(1-z^{-1})^2} = X_1(z) \cdot X_2(z)$$

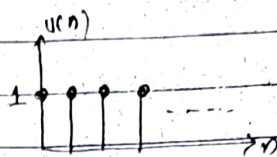
$$\text{Let, } X_1(z) = \frac{1}{1-2z^{-1}} \Rightarrow x_1(n) = (2^n)u(n)$$

$$X_2(z) = \frac{1}{(1-z^{-1})^2} \Rightarrow$$

Numericals:

Q. Find the z-transform of $x(n) = u(n)$.

So, Here, $x(n) = u(n)$



we know,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} = \sum_{n=-\infty}^{\infty} u(n) \cdot z^{-n}$$

$$= \sum_{n=-\infty}^{-1} u(n) \cdot z^{-n} + \sum_{n=0}^{\infty} u(n) \cdot z^{-n}$$

$$= 0 + \sum_{n=0}^{\infty} 1 \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^n$$

$$= (z^{-1})^0 + (z^{-1})^1 + (z^{-1})^2 + (z^{-1})^3 + \dots$$

$$\therefore X(z) = 1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 + \dots$$

we know,

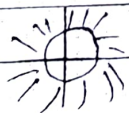
$$1 + a + a^2 + a^3 + \dots = \frac{1}{1-a}, \quad a \neq 1$$

So,

$$X(z) = \frac{1}{1 - \frac{1}{z}} \quad \text{if } \left|\frac{1}{z}\right| < 1 \quad \text{i.e. } |z^{-1}| < 1$$

$$\text{or, } 1 < |z|$$

$$\therefore X(z) = \frac{z}{z-1} \quad \text{ROC: } |z| > 1$$

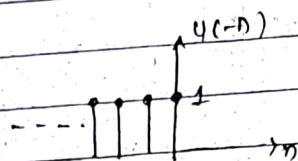


Q. Find $X(z)$ for $x(n) = u(-n)$

Solⁿ: Here, $x(n) = u(-n)$

we know,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$



$$\therefore X(z) = \sum_{n=-\infty}^0 1 \cdot z^{-n} = \sum_{n=-\infty}^0 (z^{-1})^n = \text{---}$$



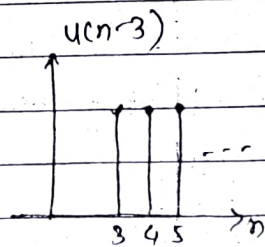
$$\therefore X(z) = \sum_{n=0}^{\infty} z^n = z^0 + z^1 + z^2 + \dots = \frac{1}{1-z} \quad \text{if } |z| < 1.$$

Q. Find $X(z)$ for $x(n) = u(n-3)$

Solⁿ: Here, $x(n) = u(n-3)$

we know,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$



$$\therefore X(z) = \sum_{n=3}^{\infty} 1 \cdot z^{-n} = \sum_{n=3}^{\infty} (z^{-1})^n$$

$$\therefore X(z) = (z^{-1})^3 + (z^{-1})^4 + (z^{-1})^5 + \dots$$

$$= (z^{-1})^3 \left[1 + (z^{-1}) + (z^{-1})^2 + (z^{-1})^3 + \dots \right]$$

$$= (z^{-1})^3 \left[\frac{1}{1-z^{-1}} \right] \quad \text{P.R.O: } |z^{-1}| < 1 \quad \text{i.e. R.O.C: } |z| > 1$$

Q. find z-transform of $x(n) = a^n u(n)$, $0 < a < 1$.

soln: we have $x(n) = a^n u(n)$

then

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n \cdot 1 \cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$$\therefore X(z) = \frac{1}{1 - az^{-1}} \quad \text{Roc: } |az^{-1}| < 1 \quad \text{i.e. Roc: } |z| > |a|$$

Q. find z-transform of $x(n) = u(n) + u(n-3) + a^n u(n)$.

soln:- we know, $X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$

Here, $x(n) = u(n) + u(n-3) + a^n u(n)$

Let $x_1(n) = u(n)$

$x_2(n) = u(n-3)$

$x_3(n) = a^n u(n)$

$\therefore X_1(z) = \frac{1}{1 - z^{-1}}$, ~~Roc~~ Roc: $|z^{-1}| < 1$ i.e. Roc $|z| > 1$

$$X_2(z) = \frac{(z^{-1})^3}{1 - z^{-1}} \quad \text{Roc: } |z^{-1}| < 1 \quad \text{i.e. Roc: } |z| > 1$$

$$X_3(z) = \frac{1}{1 - az^{-1}} \quad \text{Roc: } |az^{-1}| < 1 \quad \text{i.e. Roc: } |z| > |a|$$

finally,

$$X(z) = X_1(z) + X_2(z) + X_3(z)$$

$$= \frac{1}{1 - z^{-1}} + \frac{z^{-3}}{1 - z^{-1}} + \frac{1}{1 - az^{-1}} \quad \text{Roc: } |z| > |a|$$

2015-fall
3.a)

Q. find z-transform of the signal $x(n) = a^n \cos \omega_0 n u(n)$

Soln:-

2015-fall
3.b)

Q. find z-transform of signal $x(n) = n a^n u(n)$

Soln:- Here, given $x(n) = n a^n u(n)$ — (1)

$$\text{let } x_1(n) = a^n u(n)$$

we know,

$$\text{z-transform of } a^n u(n) \text{ is } x_1(z) = \frac{z}{z - a} \quad \text{Roc: } |z| > |a|$$

so, eqn (1) becomes

$$x(n) = n x_1(n)$$

According to differentiation property, we have

$$n x_1(n) \xleftrightarrow{z} -z \frac{d X_1(z)}{dz} \quad \text{or} \quad z^{-1} \frac{d X_1(z)}{dz^{-1}}$$

$$\therefore z \{n a^n u(n)\} = -z \frac{d}{dz} \left(\frac{z}{z-a} \right) \quad \text{or,} \quad z^{-1} \frac{d}{dz^{-1}} \left(\frac{1}{1-az^{-1}} \right)$$

$$= -z \frac{d}{dz} \left(z(z-a)^{-1} \right) \quad = +z^{-1} \cdot (-1) \cdot (-a) \cdot (1-az^{-1})^{-2}$$

$$= \frac{az^{-1}}{1}$$

$$= -z \left[z \cdot (-1) \cdot (z-a)^{-2} + (z-a)^{-1} \cdot 1 \right]$$

$$= -z \left[\frac{-z}{(z-a)^2} + \frac{1}{z-a} \right]$$

$$= -z \left[\frac{-z + (z-a)}{(z-a)^2} \right]$$

$$= -z \left[\frac{-a}{(z-a)^2} \right]$$

$$\therefore z \{n a^n u(n)\} = \frac{az}{(z-a)^2} \quad \text{ROC: } |z| > |a|$$

$$\text{OR, } z \{n a^n u(n)\} = \frac{az^{-1}}{(1-az^{-1})^2} \quad \text{ROC: } |z| > |a|$$

2016-spring
8.6)

Find the z-transform of $x(n) = -a^n u(n-1)$

Sol: Here, given, $x(n) = -a^n u(n-1)$

we have

$$Z\{u(n)\} = \frac{1}{1-z^{-1}} \quad \text{ROC: } |z| > 1$$

Applying time shifting property, we get

$$Z\{u(n-1)\} = z^{-1} \times \frac{1}{1-z^{-1}} = \frac{1}{z-1} \quad \text{ROC: } |z| > 1$$

Applying scaling property, we get

$$Z\{a^n u(n-1)\} = \frac{1}{(a^{-1}z) - 1} = \frac{a}{z-a} \quad \text{ROC: } |z| > a$$

we know, $Z\{u(-n)\} = \frac{z^{-1}}{z^{-1}-1} \quad \text{ROC: } |z^{-1}| > 1$

Applying shifting property, we get

$$Z\{u(-n-1)\} = z^{+1} \left(\frac{z^{-1}}{z^{-1}-1} \right) = \frac{z}{1-z} \quad \text{ROC: } |z| < 1$$

Applying scaling property,

$$Z \{ a^n u(-n-1) \} = \frac{(a^{-1}z)}{1 - (a^{-1}z)} \quad \text{ROC: } |a^{-1}z| < 1$$

i.e. ROC: $|z| > |a|$

$$= \frac{z}{a-z} \quad |z| < |a|$$

$$\therefore Z \{ -a^n u(-n-1) \} = \frac{-z}{a-z} \quad \text{ROC: } |z| < |a|$$

$$\text{hence, } Z \{ -a^n u(-n-1) \} = \frac{z}{z-a} \quad \text{ROC: } |z| > |a|$$

Q. Find the z-transform of $x(n) = n^2 u(n)$.

Solⁿ: Here, given, $x(n) = n^2 u(n)$
 $= n \cdot n u(n)$

we know, $Z \{ u(n) \} = \frac{1}{1-z^{-1}} \quad \text{ROC: } |z| > 1$

using differentiation property,

$$Z \{ n u(n) \} = z^{-1} \frac{d}{dz^{-1}} \left(\frac{1}{1-z^{-1}} \right)$$

$$= z^{-1} (-1) \cdot (1-z^{-1})^{-2} (-1)$$

$$= \frac{z^{-1}}{(1-z^{-1})^2} \quad \text{ROC: } |z^{-1}| < 1$$

i.e. ROC: $|z| > 1$

$$\therefore z \{n \cdot u(n)\} = \frac{z^{-1}}{(1-z^{-1})^2} \quad \text{ROC: } |z| > 1$$

Again applying differentiation property

$$z \{n^2 \cdot u(n)\} = z^{-1} \frac{d}{dz^{-1}} \left[\frac{z^{-1}}{(1-z^{-1})^2} \right]$$

$$= z^{-1} \cdot \left[\frac{1 \cdot (1-z^{-1})^2 - (2)(1-z^{-1})^2 \cdot (-1)}{\{(1-z^{-1})^2\}^2} \right]$$

$$\therefore z \{n^2 u(n)\} = z^{-1} \left[\frac{(1-z^{-1})^2 + 2(1-z^{-1})}{(1-z^{-1})^4} \right]$$

Q. Find the z-transform of $x(n) = \cos \omega_0 n u(n)$. $n > 0$

Solⁿ:- Here, $x(n) = \cos \omega_0 n u(n)$

$$= \left[\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2} \right] u(n)$$

$$x(n) = \underbrace{\frac{1}{2} e^{j\omega_0 n} u(n)}_{x_1(n)} + \underbrace{\frac{1}{2} e^{-j\omega_0 n} u(n)}_{x_2(n)}$$

$$\text{Let } x_1(n) = \frac{1}{2} e^{j\omega_0 n} u(n) \quad \& \quad x_2(n) = \frac{1}{2} e^{-j\omega_0 n} u(n)$$

$$\therefore X_1(z) = \frac{1}{2} \sum_{n=0}^{\infty} e^{j\omega_0 n} z^{-n}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (e^{j\omega_0} z^{-1})^n$$

$$= \frac{1}{2} \left[1 + e^{j\omega_0} z^{-1} + (e^{j\omega_0} z^{-1})^2 + (e^{j\omega_0} z^{-1})^3 + \dots \right]$$

$$\therefore X_1(z) = \frac{1}{2} \left[\frac{1}{1 - e^{j\omega_0} z^{-1}} \right] \quad \text{i.e. Roc: } |e^{j\omega_0} z^{-1}| < 1$$

i.e. Roc: $|z| > |e^{j\omega_0}|$

i.e. $|z| > 1$

Similarly,

$$X_2(z) = \frac{1}{2} \cdot \frac{1}{1 - e^{-j\omega_0} z^{-1}} \quad \text{Roc: } |z| > |e^{-j\omega_0}|$$

i.e. $|z| > 1$

now,

$$X(z) = X_1(z) + X_2(z)$$

$$= \frac{1}{2} \cdot \frac{1}{1 - e^{j\omega_0} z^{-1}} + \frac{1}{2} \cdot \frac{1}{1 - e^{-j\omega_0} z^{-1}} \quad \text{Roc: } |z| > 1$$

$$= \frac{1}{2} \left[\frac{(1 - e^{-j\omega_0} z^{-1}) + (1 - e^{j\omega_0} z^{-1})}{(1 - e^{j\omega_0} z^{-1})(1 - e^{-j\omega_0} z^{-1})} \right]$$

$$X(z) = \frac{1}{2} \left[\frac{2 - z^{-1} (e^{j\omega_0} + e^{-j\omega_0})}{1 - e^{-j\omega_0} z^{-1} - e^{j\omega_0} z^{-1} + e^{-2}} \right] \quad \text{Roc: } |z| > 1$$

$$= \frac{1}{2} \left[\frac{2 - z^{-1} \cdot 2 \cos \omega_0}{1 - z^{-1} (e^{j\omega_0} + e^{-j\omega_0}) + z^{-2}} \right]$$

$$= \frac{1}{2} \left[\frac{2 - 2z^{-1} \cos \omega_0}{1 - z^{-1} (2 \cos \omega_0) + z^{-2}} \right]$$

$$\therefore X(z) = \frac{1 - z^{-1} \cos \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}} \quad |z| > 1$$

(1)

Similarly, for $x(n) = \sin \omega_0 n u(n)$

$$X(z) = \mathcal{X} \{ \sin \omega_0 n u(n) \} = \frac{1 - z^{-1} \sin \omega_0}{1 - 2z^{-1} \cos \omega_0 + z^{-2}}$$

2015-fall
8.
3.2)

Find the z-transform of $x(n) = a^n \cos \omega_0 n u(n)$

Solⁿ: Applying Scaling Property in eqⁿ (1) we get

$$\begin{aligned} \mathcal{Z} \{ a^n \cos \omega_0 n u(n) \} &= \frac{1 - (a^{-1} z)^{-1} \cos \omega_0}{1 - 2(a^{-1} z)^{-1} \cos \omega_0 + (a^{-1} z)^{-2}} \\ &= \frac{1 - a z^{-1} \cos \omega_0}{1 - 2a z^{-1} \cos \omega_0 + a^2 z^{-2}}, \quad |z| > |a| \end{aligned}$$

(11)

2014-spring
2.2)

Find z-transform of $x(n) = n \cdot a^n \cos \omega_0 n \cdot u(n)$

Soln:- we have from eqn (1),

Applying differentiation property we get

$$z \{ n \cdot a^n \cos \omega_0 n \cdot u(n) \} = z^{-1} \frac{d}{dz^{-1}} \left\{ \frac{1 - a z^{-1} \cos \omega_0}{1 - 2a z^{-1} \cos \omega_0 + a^2 z^{-2}} \right\}$$

$$= z^{-1} \left[\frac{-a \sin \omega_0 (1 - 2a z^{-1} \cos \omega_0 + a^2 z^{-2}) - (-2a \cos \omega_0 + 2a^2 z^{-1}) (1 - a z^{-1})}{(1 - 2a z^{-1} \cos \omega_0 + a^2 z^{-2})^2} \right]$$

2015-spring
2.4)

A discrete time signal is given by the expression

$$x(n) = n \left(-\frac{1}{2} \right)^n u(n) \otimes \left(\frac{1}{4} \right)^{-n} u(-n). \text{ Find z-transform}$$

Soln:- Here, let, $x_1(n) = n \left(-\frac{1}{2} \right)^n$ & $x_2(n) = \left(\frac{1}{4} \right)^{-n} u(-n)$

For $x_1(n) = n \left(-\frac{1}{2} \right)^n$

we know,

$$z \{ u(n) \} = \frac{1}{1 - z^{-1}} \quad \text{ROC: } |z| > 1$$

using differentiation property

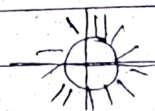
$$z \{n u(n)\} = z^{-1} \frac{d}{dz^{-1}} \left\{ \frac{1}{1-z^{-1}} \right\}$$

$$= \frac{z^{-1}}{(1-z^{-1})^2} \quad \text{ROC: } |z| > 1$$

Applying scaling property

$$z \left\{ n \left(-\frac{1}{2}\right)^n u(n) \right\} = \frac{\left\{ \left(-\frac{1}{2}\right)^{-1} z \right\}^{-1}}{\left[1 - \left\{ \left(-\frac{1}{2}\right)^{-1} z \right\}^{-1} \right]^2}$$

$$\therefore X_1(z) = \frac{\left(-\frac{1}{2}\right) z}{\left[1 + \frac{1}{2} z^{-1} \right]^2} \quad \text{ROC: } |z| > \frac{1}{2}$$



$$\text{For } x_2(n) = \left(\frac{1}{4}\right)^n u(n) = .4^n u(n)$$

we know,

$$z \{u(n)\} = \frac{1}{1-z^{-1}} \quad \text{ROC: } |z| > 1$$

using Time reversal property

$$z \{u(-n)\} = \frac{1}{1-(z^{-1})^{-1}} = \frac{1}{1-z} \quad \text{ROC: } |z| < 1$$

Applying scaling property

$$\therefore \mathcal{Z}\{4^n u(-n)\} = \frac{1}{1 - (4^{-1}z)} = \frac{1}{1 - \frac{1}{4}z} \quad \text{Roc: } |z| < 4$$

here,

$$\text{i.e. } x_2(z) = \frac{4}{4-z} \quad \text{Roc: } |z| < 4$$

$$\therefore x(n) = x_1(n) \oplus x_2(n)$$

$$\therefore X(z) = x_1(z) \cdot x_2(z)$$



$$\therefore X(z) = \frac{\left(-\frac{1}{2}\right)z^{-1}}{\left[1 + \frac{1}{2}z^{-1}\right]^2} \cdot \frac{4}{4-z} \quad \text{Roc: } \frac{1}{2} < |z| < 4$$

soln-
3.2) Q. Find z-transform and its Roc for signal.

$$x(n) = \left(-\frac{1}{3}\right)^n u(n) - u(-n-1)$$

$$\text{Soln: Let } x_1(n) = \left(-\frac{1}{3}\right)^n u(n)$$

$\therefore$ taking z-transform

$$X_1(z) = \frac{1}{1 + \frac{1}{3}z^{-1}} \quad \text{Roc: } |z| > \frac{1}{3}$$

$$\text{Let, } x_2(n) = u(-n-1)$$

$$\therefore X_2(z) = ? ; \text{ we know, } \mathcal{Z}\{u(n)\} = \frac{1}{1-z^{-1}} ; |z| > 1$$

$$u\{n+1\}$$

Applying time reversal property

$$Z\{u(-n)\} = \frac{1}{1-(z^{-1})^{-1}} = \frac{1}{1-z} \quad \text{ROC: } |z| < 1$$

Applying time shifting property.

$$\therefore X_2(z) = Z\{u(-n-1)\} = z^{-1} \left(\frac{1}{1-z} \right) = \frac{z^{-1}}{1-z} \quad |z| < 1$$

now,

$$x(n) = x_1(n) - x_2(n)$$

$$\therefore X(z) = X_1(z) - X_2(z)$$

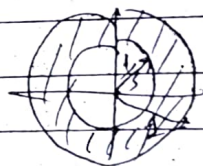
$$\therefore X(z) = \frac{1}{1 + \frac{1}{3}z^{-1}} - \frac{z^{-1}}{1-z} \quad \text{ROC: } \frac{1}{3} < |z| < 1$$

OR $u(-n-1)$

$$Z\{u(n)\} = \frac{1}{1-z^{-1}} \quad |z| > 1$$

$$= Z\{u(n+1)\} = \frac{z}{1-z^{-1}} \quad |z| > 1$$

$$X_2(z) = Z\{u(-n-1)\} = \frac{z^{-1}}{1-z} \quad |z| < 1$$



$$\therefore X(z) = X_1(z) - X_2(z)$$

$$= \frac{1}{1 + \frac{1}{3}z^{-1}} - \frac{z^{-1}}{1-z} \quad \text{ROC: } \frac{1}{3} < |z| < 1$$

2017 Spring
2.67

If the causal and stable LTI system is described by following LCCD eqn

$$3y(n) = y(n-1] + x(n-1)$$

- i) find z-transform transfer function of the system
- ii) find impulse response $h(n)$ of the system
- iii) Plot magnitude of this system

Solⁿ: Here, $3y(n) = y(n-1] + x(n-1)$

ii) taking z-transform,

$$3Y(z) = z^{-1}Y(z) + z^{-1}X(z)$$

$$\text{or } Y(z) \{ 3 - z^{-1} \} = X(z) \{ z^{-1} \}$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{3 - z^{-1}}$$

$\therefore$ z-transform transfer function of the system

$$H(z) = \frac{z^{-1}}{3 - z^{-1}} = \frac{z^{-1}}{3 - z^{-1}} \quad \text{ROC: } |z^{-1}| < 3; |z| > 3$$

$$\text{i.e. } H(z) = \frac{1}{3} \cdot \frac{z^{-1}}{1 - \frac{1}{3}z^{-1}} \quad \therefore H$$

ii) taking inverse z-transform of $H(z)$ we get

where

$$H(z) = z^{-1} \cdot \frac{1}{3} \cdot \frac{1}{1 - \frac{1}{3}z^{-1}}$$

$$\therefore h(n) = \frac{1}{3} \left(\frac{1}{3} \right)^{n-1} u(n-1) = \left(\frac{1}{3} \right)^n u(n-1)$$

2016-fall
82)

Q. find the inverse z -transform of the system $X(z) = \frac{5-2z^{-1}}{2-z^{-1}-1.5z^{-2}}$

for the ROC: (i) $|z| < 1$ (ii) $|z| > 3$ (iii) $1 < |z| < 3$.

Solⁿ: - here, $X(z) = \frac{5-2z^{-1}}{2-z^{-1}-1.5z^{-2}}$

$$\begin{aligned} & \text{or } X(z) = \frac{z^{-1}(5z-2)}{z^{-2}(2z^2-z-1.5)} \\ & = \frac{5z-2}{2z^2-z-1.5} \quad X(z) = \frac{z(5z-2)}{2z^2-z-1.5} \\ & = \frac{5z-2}{2z^2-z-1.5} \quad \text{now, dividing both sides by } z \\ & = \frac{5z-2}{2z^2-z-1.5} \quad \frac{X(z)}{z} = \frac{z(5z-2)}{z(2z^2-z-1.5)} \\ & = \frac{5z-2}{2z^2-z-1.5} \quad \therefore X(z) = \frac{5z-2}{2z^2-z-1.5} \end{aligned}$$

Let, $X(z) = \frac{5z-2}{2z^2-z-1.5}$

2014 fall
u.b

An LTI System is characterized by the system function

$$H(z) = \frac{3-4z^{-1}}{1-\frac{7}{2}z^{-1}+\frac{3}{2}z^{-2}}$$

Specify ROC of $H(z)$ & find $h(n)$.

When the System is stable?

Soln:-

Here, given,

$$H(z) = \frac{3-4z^{-1}}{1-\frac{7}{2}z^{-1}+\frac{3}{2}z^{-2}} = \frac{z^{-1}(3z-4)}{z^{-2}(1z^2-\frac{7}{2}z+\frac{3}{2})}$$

$$\therefore H(z) = \frac{3z-4}{z^2-\frac{7}{2}z+\frac{3}{2}}$$

2013-spring
2.6)

Q. The z-transform of a particular discrete-time signal $x(n)$ is expressed as $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$. Determine $x(n)$ using time shifting property.

Soln: Here, given, $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$

taking inverse z-transform

$$x(n) = z^{-1} X(z) = z^{-1} \left[\frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \right]$$

$$\text{or, } x(n) = z^{-1} \left[\frac{1}{1 - \frac{1}{2}z^{-1}} \right] + z^{-1} \left[\frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \right]$$

$$x(n) = z^{-1} [X_1(z)] + z^{-1} [X_2(z)] \quad \text{--- (1)}$$

for where, $X_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$	& $X_2(z) = \frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$
---	--

$$X_1(z) = z \left\{ \left(\frac{1}{2} \right)^n \cdot u(n) \right\}$$

$$X_2(z) = \frac{1}{2} \cdot z^{-1} X_1(z)$$

we know

Applying ^{time} Shifting Property i.e. $x(n) \leftrightarrow X(z)$

in $X_2(z)$

$$x(n-k) \leftrightarrow z^{-k} X(z)$$

$$z^{-1} (X_2(z)) = \frac{1}{2} z^{-1} X_1(z) =$$

$$= \frac{1}{2} z \left\{ \left(\frac{1}{2} \right)^{n-1} u(n-1) \right\}$$

Substituting $X_1(z)$ & $X_2(z)$ in eqn (1)

$$x(n) = z^{-1} \left\{ z \left(\frac{1}{2}\right)^n \cdot u(n) \right\} + z^{-1} \left\{ \frac{1}{2} z \left\{ u(n-1) \right\} \left(\frac{1}{2}\right)^{n-1} \right\}$$

$$= \left(\frac{1}{2}\right)^n u(n) + \frac{1}{2} \left\{ u(n-1) \left(\frac{1}{2}\right)^{n-1} \right\}$$

$$= \left(\frac{1}{2}\right)^n \{ u(n) + u(n-1) \}$$

$$= \left(\frac{1}{2}\right)^n \{ u(n) - u(n-1) + 2u(n-1) \}$$

$$\therefore x(n) = \left(\frac{1}{2}\right)^n \{ \delta(n) + 2u(n-1) \} \quad \text{Q}$$

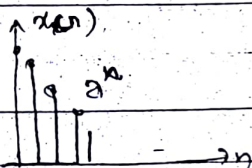
2017-18
3.2

Given, that $x(n) = a^n u(n)$. determine one-sided z-transform of a signal $x_1(n) = x(n+2)$

Sol: Here given, $x(n) = a^n u(n)$

We know,

$$x(z) = z \left\{ a^n u(n) \right\} = \frac{1}{1 - az^{-1}} \quad \text{ROC: } |z| > |a|$$



but we have to calculate one-sided z-transform of $x_1(n) = x(n+2)$.
Since $u(n)$ is one-sided transform.

$$\therefore X_1(z) = z \{ x_1(n) \} = z^2 x(z) = z^2 \left(\frac{1}{1 - az^{-1}} \right) \quad \text{ROC: } |z| > |a|$$

We know, from time advance, property $x(n+k) \xrightarrow{z^k} z^k \left[x^+(z) - \sum_{n=0}^{k-1} x(n) z^{-n} \right]$

Here, $k=2$ so,

$$\therefore z^2 [x(n+2)] = z^2 \left[x^+(z) - \sum_{n=0}^{2-1} x(n) \cdot z^{-n} \right] = z^2 x^+(z) - x(0) z^2 - x(1) z$$

$$\text{but } x(0) = 1, x(1) = a \text{ and } x^+(z) = \frac{1}{1 - az^{-1}}$$

$$\therefore z^2 [x(n+2)] = \frac{z^2}{1 - az^{-1}} - z^2 - az \quad \text{Q}$$

2012-fall
3.2)

If the z-transform of any signal is $X(z) = 2 + 3z^{-1} + 4z^{-2}$
Determine the initial & final values of the corresponding discrete-time signal $x(n)$.

Solⁿ: Here,

$$X(z) = 2 + 3z^{-1} + 4z^{-2}$$

from initial value theorem

$$x(0) = \lim_{n \rightarrow 0} x(n) = \lim_{|z| \rightarrow \infty} X(z)$$

$$\therefore x(0) = \lim_{|z| \rightarrow \infty} (2 + 3z^{-1} + 4z^{-2}) = \lim_{z \rightarrow \infty} \left(2 + \frac{3}{z} + \frac{4}{z^2} \right) = 2 + \frac{3}{\infty} + \frac{4}{\infty^2}$$

$$\therefore x(0) = 2 + 0 + 0 = 2$$

from final value theorem,

$$x(\infty) = \lim_{n \rightarrow \infty} x(n) = \lim_{|z| \rightarrow 1} \left[(1 - z^{-1}) X(z) \right]$$

$$\therefore x(\infty) = \lim_{|z| \rightarrow 1} \left[(1 - z^{-1}) (2 + 3z^{-1} + 4z^{-2}) \right]$$

$$= \lim_{|z| \rightarrow 1} \left[2 + 2z^{-1} + z^{-2} - 4z^{-3} \right]$$

$$= \lim_{|z| \rightarrow 1} \left[2 + \frac{2}{z} + \frac{1}{z^2} - \frac{4}{z^3} \right]$$

$$= 2 + \frac{2}{1} + \frac{1}{1} - \frac{4}{1}$$
$$= 2 + 2 - 4$$

$$\therefore x(\infty) = 0$$

$$\therefore x(0) = 2$$

$$\& x(\infty) = 0$$

PC

2013 Spring
9.6)

The z-transform of a particular discrete-time signal $x(n)$ is expressed as $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$

Determine $x(n)$ using time shifting property

Solⁿ Here, $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$

using differentiation property

$$\frac{dX(z)}{dz^{-1}} = z^{-1} \frac{d}{dz^{-1}} \left\{ \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \right\}$$

$$= z^{-1} \left\{ \frac{\frac{1}{2}(1 - \frac{1}{2}z^{-1}) - (-\frac{1}{2})(1 + \frac{1}{2}z^{-1})}{(1 - \frac{1}{2}z^{-1})^2} \right\}$$

$$= z^{-1} \left\{ \frac{\frac{1}{2} - \frac{1}{4}z^{-1} + \frac{1}{2} + \frac{1}{4}z^{-1}}{(1 - \frac{1}{2}z^{-1})^2} \right\}$$

$$= z^{-1} \left\{ \frac{1 + 0}{1 - \frac{1}{2}z^{-1}} \right\}$$

$$\frac{dX(z)}{dz^{-1}} = z^{-1} \cdot \frac{1}{1 - \frac{1}{2}z^{-1}}$$

multiplying both sides by z^{-1}

$$z^{-1} \frac{dX(z)}{dz^{-1}} = z^{-1} \cdot \left\{ \frac{z^{-1}}{1 - \frac{1}{2}z^{-1}} \right\}$$

$$n x(n) = z^{-2} \cdot \left(\frac{1}{2}\right)^n u(n)$$

$$\therefore n x(n) = \left(\frac{1}{2}\right)^n u(n-2)$$

$$\therefore x(n) = \frac{\left(\frac{1}{2}\right)^n u(n-2)}{n}$$

Q. 5.2)

Given, $y(n) = 3y(n-1) - 0.1y(n-2) + x(n)$

- i) Find the System function, $H(z)$ iv). plot magnitude response
 ii) Is the System is IIR or FIR.
 iii) Plot pole-zero diagram.

Soln Here, $y(n) = 3y(n-1) - 0.1y(n-2) + x(n)$

taking z-transform,

$$Y(z) = 3z^{-1}Y(z) - 0.1z^{-2}Y(z) + X(z)$$

$$\therefore Y(z) \{ 1 - 3z^{-1} + 0.1z^{-2} \} = X(z)$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - 3z^{-1} + 0.1z^{-2}}$$

ii) $\Rightarrow$ Since, $H(z)$ has. ~~only~~ poles & zero. So, it is IIR System

$$H(z) = \frac{1}{z^{-2} (z^2 - 3z + 0.1)} = \frac{z^2}{z^2 - 3z + 0.1}$$

zeros lies at $z = 0$ & ∞ .

Poles lies at $z = 2.96$ & 0.0337



Q. If $X(z) = 2 + 3z^{-1} + 4z^{-2}$ then $x(n) = ?$

Solⁿ: $x(n) = \{2, 3, 4\}$

2013 Fall
2.4)

By differentiation $X(z)$ and using appropriate properties of z-transform, determine $x(n)$ for the transform given as

$$X(z) = \log(1 - z^{-1}), \text{ Roc: } |z| < \frac{1}{2}$$

L(1)

Solⁿ:- According to question

differentiate both sides in eqⁿ(1), w.r.t. z^{-1} we get

$$\frac{dX(z)}{dz^{-1}} = \frac{d \log(1 - z^{-1})}{dz^{-1}}$$

$$= \frac{1}{(1 - z^{-1})} \cdot (-1)$$

$$= \frac{-1}{1 - z^{-1}}$$

multiplying both sides by z^{-1} we get

$$z^{-1} \frac{dX(z)}{dz^{-1}} = z^{-1} \left(\frac{-1}{1 - z^{-1}} \right)$$

$$nX(n) = -1 \cdot z^{-1} \cdot \frac{1}{1 - z^{-1}}$$

we know, that

$$Z\{u(n)\} = \frac{1}{1-z^{-1}}, \quad \text{ROC: } |z| < 1 \text{ i.e. } |z| > 1$$

$$\text{so, } n x(n) = -z^{-1} u(n) = -u(n-1)$$

$$\therefore x(n) = \frac{-u(n-1)}{n}$$

2013-Spring
3.2)

The input response for a discrete-time ~~signal~~ system is given as $h(n) = \{1, 2, 3\}$ and output response is given as $y(n) = \{1, 1, 2, -1, 3\}$. Determine the $x(n)$ using ^{inverse} Z-transform.

Solⁿ: Here, given, $h(n) = \{1, 2, 3\}$ then, $H(z) = 1 + 2z^{-1} + 3z^{-2}$

$$\& y(n) = \{1, 1, 2, -1, 3\} = 1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}$$

then,

$$X(z) = ? \quad \& \quad x(n) = ?$$

Using long division Method for inverse Z-transform

$$\text{we know, } y(n) = x(n) \otimes h(n)$$

$$\therefore Y(z) = X(z) \cdot H(z)$$

$$\therefore X(z) = \frac{Y(z)}{H(z)} = \frac{1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}}{1 + 2z^{-1} + 3z^{-2}}$$

1e

$$\begin{array}{r} 1 - z^{-1} + z^{-2} \\ 1 + 2z^{-1} + 3z^{-2} \end{array} \bigg) 1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}$$

$$- 1 + 2z^{-1} + 3z^{-2}$$

$$- z^{-1} - z^{-2} - z^{-3} \quad \text{+ B/E/W}$$

$$- z^{-1} - 2z^{-2} - 3z^{-3}$$

$$+ \quad + \quad +$$

$$z^{-2} + 2z^{-3} + 3z^{-4}$$

$$z^{-2} + 2z^{-3} + 3z^{-4}$$

$$\begin{array}{ccc} - & - & - \\ x & x & x \end{array}$$

$$\therefore X(z) = 1 - z^{-1} + z^{-2} \quad \therefore x(n) = \{1, -1, 1\} \quad R$$

20/11/20
3b)

$$\text{Find } x(n) \text{ using inverse } z\text{-transform of } X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

when $\text{Roc: } |z| > 1$ and $\text{Roc: } |z| < \frac{1}{2}$.

Soln:- i) For $\text{Roc: } |z| > 1$

$$\text{Here, } X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \frac{3}{6}z^{-4} + \dots$$

$$\left(1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}\right) \cdot 1$$

must be in
descending order of power
for ROC: $|z| > 1$

$$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}$$

$$\frac{3}{2}z^{-1} - \frac{1}{2}z^{-2}$$

$$\frac{3}{2}z^{-1} - \frac{9}{4}z^{-2} + \frac{3}{4}z^{-3}$$

$$\frac{7}{4}z^{-2} - \frac{3}{4}z^{-3}$$

$$\frac{7}{4}z^{-2} - \frac{21}{8}z^{-3} + \frac{7}{8}z^{-4}$$

$$\frac{15}{8}z^{-3} - \frac{7}{8}z^{-4}$$

$$\frac{15}{8}z^{-3} - \frac{45}{16}z^{-4} + \frac{15}{16}z^{-5}$$

$$\frac{3}{6}z^{-4} - \frac{15}{16}z^{-5}$$

$$\therefore X(z) = 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \dots$$

taking inverse z-transform we get

$$x(n) = \left\{ 1, \frac{3}{2}, \frac{7}{4}, \frac{15}{8}, \dots \right\} \quad R$$

causal (and infinite)



ROC: $|z| > 1$

ii) for ROC: $|z| < \frac{1}{2}$.

$$2z^2 + 6z^3 + 14z^4 + 30z^5 + \dots$$

$$\left(\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 \right)$$

$$\frac{2z^2 + 6z^3 + 14z^4 + 30z^5 + \dots}{2z^2 - 3z + 2}$$

$$1 - 3z + 2z^2$$

$$- + -$$

$$3z - 2z^2$$

$$3z - 9z^2 + 6z^3$$

$$- + -$$

$$7z^2 - 6z^3$$

$$7z^2 - 21z^3 + 14z^4$$

$$- + -$$

$$15z^3 - 14z^4$$

$$15z^3 - 45z^4 + 30z^5$$

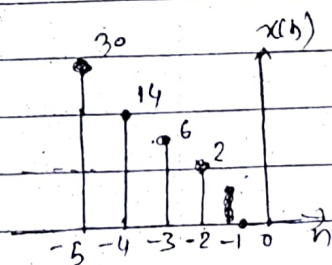
$$- + -$$

$$36z^4 - 30z^5$$

$$\therefore X(z) = 2z^2 + 6z^3 + 14z^4 + 30z^5 + \dots$$

taking inverse z -transform,

$$x(n) = \{ \dots, 30, 14, 6, 2, 0, 0 \}$$



Antircausal (∞)
& infinite.



Q: Find the inverse - z - transform of $H(z) = \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}}$

Soln: By partial fraction method.

$$\text{Here, } H(z) = \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}} = \frac{-4 + 8z^{-1}}{(1+2z^{-1})(1+4z^{-1})}$$

$$\text{Let, } \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}} = \frac{-4 + 8z^{-1}}{(1+2z^{-1})(1+4z^{-1})} = \frac{A}{1+2z^{-1}} + \frac{B}{1+4z^{-1}}$$

$$\text{or, } -4 + 8z^{-1} = A(1+4z^{-1}) + B(1+2z^{-1})$$

$$\text{or } -4 + 8z^{-1} = A + 4Az^{-1} + B + 2Bz^{-1}$$

$$\text{or } -4 + 8z^{-1} = (4A + 2B)z^{-1} + (A + B)$$

Comparing ~~the~~ Equating the coeff. of z^{-1} we get

$$4A + 2B = 8 \quad \text{--- (I)}$$

$$A + B = -4 \quad \text{--- (II)}$$

on Solving eqn (I) & (II), we get

$$A = 8 \quad \& \quad B = -12$$

$$\therefore H(z) = \frac{8}{1+2z^{-1}} + \frac{-12}{1+4z^{-1}}$$

$$\therefore H(z) = 8 \cdot \left[\frac{1}{1 - (-2z^{-1})} \right] - 12 \left[\frac{1}{1 - (-4z^{-1})} \right]$$

taking inverse - z - transform

$$\therefore h(n) = 8(-2)^n u(n) - 12(-4)^n u(n) \quad P2$$

Q. Find inverse z-transform of $X(z) = \frac{z^3}{(z+1)(z-1)^2}$ i.e. $x(n)$?

Sol. By partial fraction method:

$$\text{Here, } X(z) = \frac{z^3}{(z+1)(z-1)^2}$$

Here, the power of num. & den. are same so divide both sides by z . we get

$$\frac{X(z)}{z} = \frac{1}{z} \cdot \frac{z^3}{(z+1)(z-1)^2} = \frac{z^2}{(z+1)(z-1)^2}$$

$$\text{Let, } \frac{z^2}{(z+1)(z-1)^2} = \frac{A}{z+1} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

$$\text{or } z^2 = A(z-1)^2 + B(z+1)(z-1) + C(z+1)$$

$$\text{or } z^2 = Az^2 - 2Az + A + Bz^2 - B + Cz + C$$

$$\text{or } z^2 = (A+B)z^2 + (-2A+C)z + (A+C-B)$$

Equating the coeff. of z we get

$$A+B=1 \quad \text{--- (i)}$$

$$-2A+C=0 \quad \text{--- (ii)}$$

$$A+B+C=0 \quad \text{--- (iii)}$$

on solving we get

$$A = \frac{1}{4}$$

$$B = \frac{3}{4}$$

$$C = \frac{1}{2}$$

$$\therefore X(z) = \frac{1/4}{z+1} + \frac{3/4}{z-1} + \frac{1/2}{(z-1)^2}$$

$$\therefore X(z) = \frac{(1/4)z}{z+1} + \frac{(3/4)z}{z-1} + \frac{(1/2)z}{(z-1)^2}$$

taking inverse z-transform

$$x(n) = \frac{1}{4} (-1)^n u(n) + \frac{3}{4} (1)^n u(n) + \frac{1}{4} (1)^n n \cdot u(n)$$

2015 Spring
Q. 3.2)

Determine the inverse z-transform of

$$X(z) = \frac{z}{3z^2 - 4z + 1} \quad \text{if ROC are}$$

$$\text{i) } |z| > 1 \quad \text{ii) } |z| < \frac{1}{3} \quad \text{iii) } \frac{1}{3} < |z| < 1$$

Solⁿ: use partial fraction method not long division method

$$\text{Here, } X(z) = \frac{z}{3z^2 - 4z + 1}$$

(always) Dividing both sides by 'z' we get

$$\frac{X(z)}{z} = \frac{1}{z(3z^2 - 4z + 1)}$$

$$= \frac{1}{3z^2 - 4z + 1}$$

$$X(z) = \frac{1}{z(3z^2 - 3z - 2 + 1)} = \frac{1}{(z-1)(3z-1)}$$

$$\text{Let, } \frac{1}{(z-1)(3z-1)} = \frac{A}{z-1} + \frac{1}{3z-1}$$

$$A = \frac{1}{3z-1} \Big|_{z=1} = \frac{1}{2}$$

$$B = \frac{1}{z-1} \Big|_{z=\frac{1}{3}} = -\frac{3}{2}$$

$$\therefore X(z) = \frac{\frac{1}{2}}{z-1} - \frac{\frac{3}{2}}{3z-1}$$

$$\therefore X(z) = \frac{1}{2} \left(\frac{z}{z-1} \right) - \frac{1}{2} \frac{z}{(z-\frac{1}{3})}$$

i) For $|z| > 1$,

$$x(n) = \frac{1}{2} (1)^n u(n) - \frac{1}{2} \left(\frac{1}{3}\right)^n u(n)$$

$\uparrow$ causal $\uparrow$ causal

ii) For $|z| < \frac{1}{3}$

$$x(n) = \frac{1}{2} (1)^n u(-n-1) - \frac{1}{2} \left(\frac{1}{3}\right)^n u(-n-1)$$

$$\text{or } x(n) = \frac{1}{2} (1)^{-n-1} u(-n-1) - \frac{1}{2} \left(\frac{1}{3}\right)^{n-1} u(-n-1)$$

$\underbrace{\hspace{10em}}$
 both non-causal

iii) For $\frac{1}{3} < |z| < 1$

$$x(n) = \underbrace{\frac{1}{2}(1)^n u(-n-1)}_{\text{non-causal}} - \underbrace{\frac{1}{2}\left(\frac{1}{3}\right)^n u(n)}_{\text{causal}}$$

2016-Spring

Q. The z transform of particular discrete signal $x(n]$ is expressed as

$$X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{3}z^{-1}} \quad \text{determine } x(n)$$

Soln:- By using long division method.

$$\begin{array}{r} 1 + z^{-1} + \frac{1}{2}z^{-2} + \dots \\ 1 - \frac{1}{3}z^{-1} \overline{) 1 + \frac{1}{2}z^{-1}} \\ \underline{1 - \frac{1}{3}z^{-1}} \\ z^{-1} - \frac{1}{2}z^{-2} \\ \underline{z^{-1} - \frac{1}{2}z^{-2}} \\ \phantom{z^{-1}} + \frac{1}{2}z^{-2} \\ \underline{\phantom{z^{-1}} + \frac{1}{2}z^{-2}} \\ \phantom{z^{-1}} \phantom{+ \frac{1}{2}z^{-2}} - \frac{1}{4}z^{-3} \end{array}$$